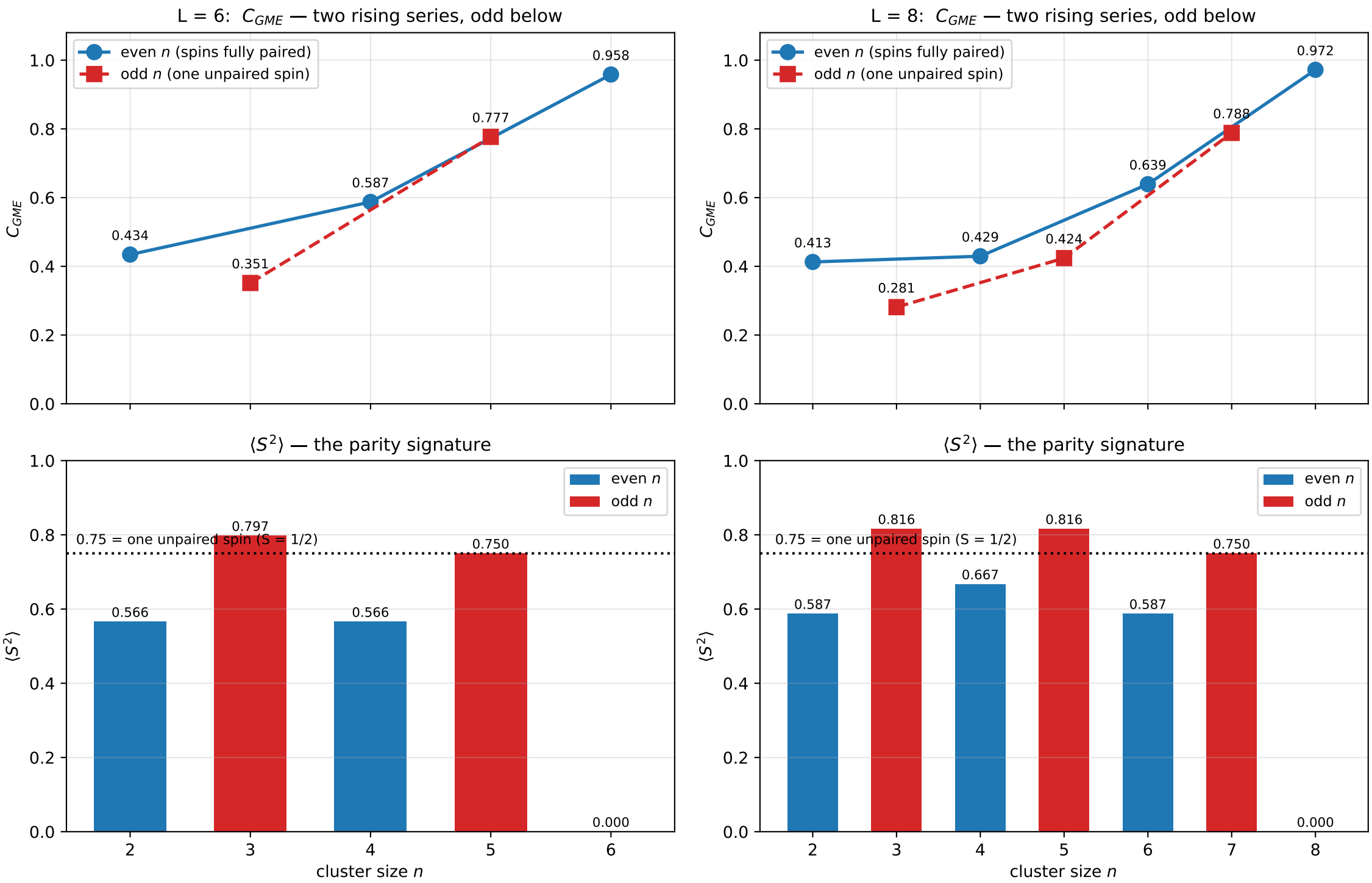
Why C_{GME} is not monotonic in n: an even/odd (parity) effect — Hubbard ring, $U = 10$, $t = 0.05$, $T = 0$
L = 6: C_{GME} — two rising series, odd below
even n (spins fully paired)
odd n (one unpaired spin)
C_{GME}
0.434
0.351
0.587
0.777
0.958
L = 8: C_{GME} — two rising series, odd below
0.413
0.281
0.429
0.424
0.639
0.788
0.972
$\langle S^2 \rangle$ — the parity signature
even n
odd n
0.75 = one unpaired spin (S = 1/2)
$\langle S^2 \rangle$
0.566
0.797
0.566
0.750
0.000
0.587
0.816
0.667
0.816
0.587
0.750
0.000
cluster size n